

Alpilla 600 MW Combined Cycle Power Plant

Document numbering and document control procedure

Document number	ALP-EPC-00000-GE-PRC-0002
Revision	draft - not yet issued (next issue: IFR rev A)
Originator	Istanbul EPC
Review class	approval
Planned issue (workflow timeline)	IFR 2026-12-21, IFA 2027-01-18, IFC 2027-01-25
Basis	requirement:ER-17.01

1. Purpose, scope and basis

This procedure sets the numbering of all project documents (consortium, Imaginary Electric, subsuppliers), the document types and their review class, revisions and purposes, the review cycle with the Owner and the maintenance of the master document list (MDL).

ER-17.01: Documents submitted electronically with the project numbering (consortium proposal adopted by the Owner in TQ-035: ALP-<ORG>-<KKS>-<DISC>-<TYPE>-<NNNN>) and a master document register; review period 14 days (21 days for the Basic Design Package).

- TQ-035 (closed, answered 2026-09-28 by Owner): (1) Accepted: the consortium numbering is the project numbering under ER-17.01; the Owner's EDMS will use it unchanged. (2) The KKS key list is approved for engineering. The Owner's Engineer verifies each key title against the VGB-S-811 key catalogue with the KKS manual review; until then keys stay flagged 'not verified'. (3) MDL accepted as baseline for the Basic Design Package. Review classes accepted with one change: logic diagrams (LOG) incl. cause and effect are safety relevant and are for approval. The MDL is updated monthly with the progress report. (4) AWP definitions accepted. The Owner requires every CWP to be released for construction only when its EWP is complete (all IFC) and its PWP materials are on site or have a confirmed delivery before the planned installation.
- TQ-IEC-023 (closed, answered 2026-09-28 by Imaginary Electric): (1) Confirmed. IEC tags the power island with the project KKS to component level (B), systems and equipment units per the project key list; IEC part numbers are kept as vendor attributes below component level. IEC checks the key titles of its main groups against its VGB-S-811 licence copy and reports deviations with the KKS manual review (IFR). (2) Confirmed: project number primary, IEC number as vendor reference on the title block. (3) Confirmed: the VDRL dates in the MDL are the IEC commitment. (4) Confirmed with an improvement: generator shop tests are brought forward; the GT generators leave the works in week 49 together with the GT (sea transport 4 weeks), on site 2027-11-08. The ST generator follows the ST train (no change).

2. Document number

Format: ALP-<ORG>-<KKS>-<DISC>-<TYPE>-<NNNN>

Field	Format	Meaning
ALP	fixed	Project code
ORG	3 characters	Originator (section 3)
KKS	NNAAA / 00000	KKS system G F0 F1F2F3 of the main system the document belongs to (ALP-EPC-00000-GE-PRC-0001); 00000 for plant-general documents
DISC	2 letters	Discipline (section 3)
TYPE	3 characters	Document type (section 4); defines the IEC 61355 class and the workflow rules
NNNN	4 digits	Running number per ORG-KKS-DISC-TYPE, from 0001; never reused

Example ALP-EPC-00ACA-EL-DSH-0001: originator EPC (Istanbul EPC (consortium leader)); KKS 00ACA (380 kV switchgear (GIS)); discipline EL (electrical); type DSH; number 0001

Sheets of a multi-sheet drawing share the number (sheet count in the MDL); a supplier's own number is recorded as vendor reference and shown as a secondary number in the title block (TQ-IEC-023).

3. Originator and discipline codes

ORG	Originator
EPC	Istanbul EPC (consortium leader)
IEC	Imaginary Electric (consortium member)

ORG	Originator
OWN	Owner
xxx	Subsupplier: 3-character code allocated by Istanbul EPC with the purchase order

DISC	Discipline
GE	general
PR	process
ME	mechanical
PI	piping
EL	electrical
IC	instrumentation and control
CV	civil
SS	structural
HV	hvac
HS	hse
CM	commissioning
PC	project control

4. Document types and workflow rules

Prep = working days from inputs ready to first issue (longer for multi-sheet documents, at most double); Update = working days to incorporate comments; Inputs at = maturity the inputs must have reached (IFR or IFC); C = construction document (belongs to an EWP), P = procurement document (belongs to an MR).

Type	Title	IEC 61355	Review class	Prep	Upd.	Inputs at	Typical inputs	C/P
ARC	Architectural drawing	L	approval	20	10	IFR	LAY	C
BLK	Block / architecture diagram	F	approval	15	10	IFR	DBR	
CAL	Calculation	E	review	15	5	IFR	SDS, SLD, STU, PID, DBR, CAL, HMB	
CBL	Cable schedule and routing	M	information	15	5	IFR	SLD, LAY	C
CMP	Commissioning procedure	B	approval	20	10	IFR	PID, CNR, CAL, STU, SPC	
CNR	Control narrative / functional description	F	review	15	10	IFR	PID, CNR	
CVL	Civil drawing (grading, roads, drainage, marine)	L	approval	20	10	IFR	PLT, STU, SDS	C
DBR	Design basis report	D	approval	20	10	IFR	-	
DSH	Datasheet	E	approval	10	5	IFR	SPC	P
ELY	Electrical layout (earthing, lighting, trays, equipment)	L	review	20	10	IFR	LAY, SLD	C
FND	Foundation / underground drawing	L	approval	20	10	IFR	LAY, CAL	C
GAD	General arrangement drawing	L	review	20	10	IFR	LAY	C
HAZ	Safety study (HAZID, HAZOP, SIL, hazardous areas, fire, QRA)	S	approval	20	10	IFR	PID, PLT, HAZ	
HMB	Heat and mass balance	F	approval	15	10	IFR	DBR	
IOL	I/O list	P	review	15	10	IFR	PID, CNR, LST, IOL	
ISO	Isometric drawing package	L	information	20	5	IFR	PPL	C
ITP	Inspection and test plan	Q	approval	10	5	IFR	SPC	
LAY	Equipment layout / arrangement	L	approval	20	10	IFR	PLT	C
LOG	Logic diagram	F	approval	20	10	IFR	CNR, IOL	
LOP	Loop / connection diagram	M	information	15	5	IFR	IOL	C
LST	List / index (equipment, lines, valves, instruments, loads, tie-ins)	P	review	15	10	IFR	PID, PFD, LST, SLD, DBR, PRC	
M3D	3D model review report (30 / 60 / 90 %)	L	review	15	5	IFR	PLT, LAY, PPL, ISO	
MDL	Master document list	A	information	10	5	IFR	-	
MRQ	Material requisition	C	information	5	3	IFR	SPC	P
OMM	Operation and maintenance manual	D	review	30	10	IFR	PID	
PFD	Process flow diagram	F	approval	15	10	IFR	HMB	
PID	Piping and instrumentation diagram	F	approval	25	15	IFR	SDS	
PLN	Plan (execution, quality, HSE, commissioning, training)	B	approval	20	10	IFR	PRC, PLN, DBR, PFD, LST	
PLT	Plot plan	L	approval	20	10	IFR	DBR	
PPL	Piping general arrangement / plans	L	review	25	10	IFR	PID, LAY	C
PRC	Procedure / project standard	B	approval	15	10	IFR	-	
RPT	Technical report	D	review	15	5	IFR	-	
SCH	Schedule	B	review	15	5	IFR	MDL	
SDS	System design description	D	review	15	10	IFR	DBR	
SLD	Single line diagram	F	approval	15	10	IFR	LST	
SPC	Technical specification	E	approval	20	10	IFR	-	P
STL	Steel structure drawing	L	approval	20	10	IFR	GAD, CAL	C
STR	Pipe stress analysis report	E	review	20	5	IFR	PPL	
STU	Engineering study	D	review	25	10	IFR	-	

5. Number allocation

- Numbers are allocated only in the database: `python -m engine kks --next <ORG> <KKS> <DISC> <TYPE>` gives the next free number; the document is registered with `python -m engine db add document ... --reason`.
- Every document record carries `type_code`, discipline, originator and system consistent with its number; the `engineering_plan` engine checks all numbers at every run.
- A cancelled document keeps its number (status cancelled, purpose VOID); numbers are never reused.

6. Revisions and issue purposes

Purpose	Meaning	Revision
IFR	Issued for review / first issue	A, B, C ...
IFA	Issued for approval (approval class, after the Owner's comments are incorporated)	A, B, C ...
IFD	Issued for design (released internally for dependent design, e.g. vendor data frozen)	A, B, C ...
IFP	Issued for purchase (requisitions, specifications, datasheets)	A, B, C ...
IFC	Issued for construction / final	0, 1, 2 ...
IFI	Issued for information (information class, supplier documents received)	A, B, C ... or supplier rev.
AB	As built	next number
VOID	Cancelled; the number is never reused	-

Each issue is a `document_revision` record (revision, purpose, date, prepared / checked / approved by, file); packages are transmitted with `python -m engine deliver` (frozen copy and transmittal).

7. Review and approval

Class	Cycle
approval	Owner approval: IFR -> Owner review -> update -> IFA -> approval code 1 -> IFC
review	Owner review: IFR -> Owner review -> update -> IFC
information	sent to the Owner for information: IFR -> update -> IFC (no review period)
internal	consortium internal, not transmitted for review

Owner review period: 10 working days (14 calendar days, ER-17.01), 15 working days (21 calendar days) for Basic Design Package documents; after approval code 1 the IFC follows within 5 working days. Review codes:

Code	Meaning
1	No comments - proceed (approval: approved)
2	Proceed with the comments incorporated; re-issue
3	Do not proceed; revise and re-submit
4	For information / reference only; no review

Comments are recorded as `review_comment` records against the revision (comment -> response -> closed in the revision that incorporates it). The review class of a document type may be overridden per document (review field of the MDL line).

8. Supplier documents

- Imaginary Electric and subsuppliers submit per their VDRL, which is part of the MDL (originator IEC / subsupplier code) with committed first-issue and final dates.
- Supplier documents are numbered by Istanbul EPC before the purchase order, carry the project number as the primary number and are reviewed by Istanbul EPC before transmittal to the Owner where the class requires it.
- Received files are registered as source records (sources/) and linked to the MDL line (source field) with an IFR / IFI revision.

9. Master document list

The MDL is generated by the `engineering_plan` engine from the document records (623 documents: EPC 586, IEC 37; 22 in the Basic Design Package). It is updated with every change of the records and issued monthly with the progress report (TQ-035).

Discipline	Documents	Hours	Sheets
civil	42	3,072	140
commissioning	11	960	11
electrical	82	5,215	144
general	15	1,440	15
hse	16	2,420	21
hvac	11	871	21
i and c	88	6,020	264
mechanical	155	6,620	155

Discipline	Documents	Hours	Sheets
pipng	89	11,556	983
process	98	9,825	198
project control	2	160	2
structural	14	1,158	65

Check results (engine)

1 OK, 0 INFO, 0 WARN (checks: numbering). WARN must be resolved before the next issue; INFO is noted.

Item	Check	Result	Detail
-	numbering	OK	623 document numbers checked, 0 finding(s)